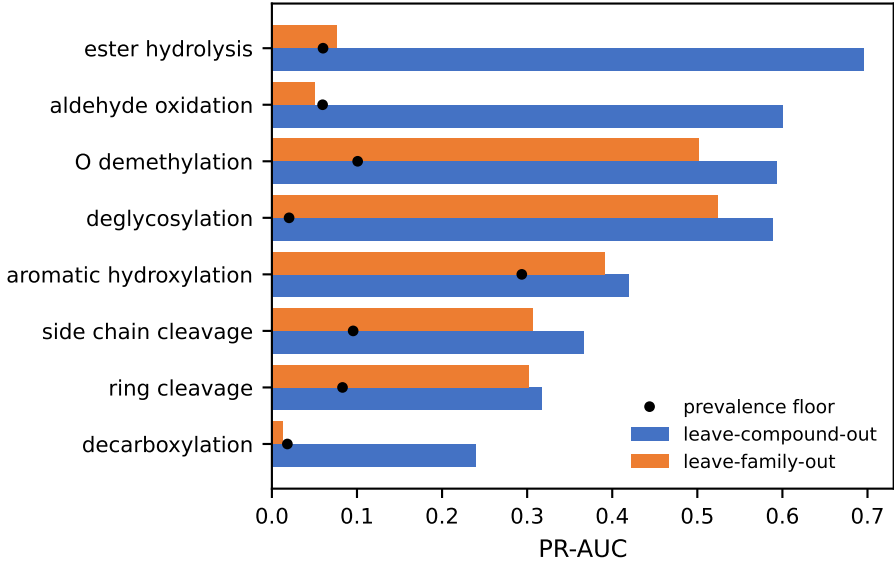


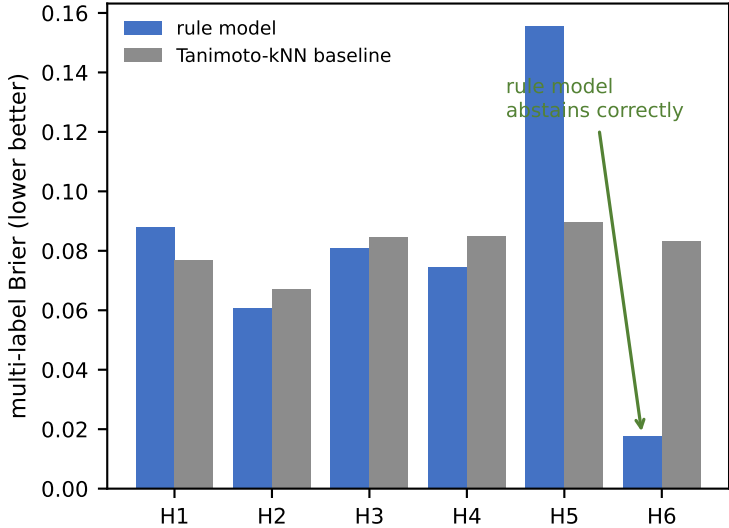
SEAGRAMMAR — in silico pilot on the final 20 + 6 compound panel

Rule-learning framework (WP0) run end-to-end on the finalised panel: 20 training compounds, 6 sealed compounds, 42 strains, 3 replicates.

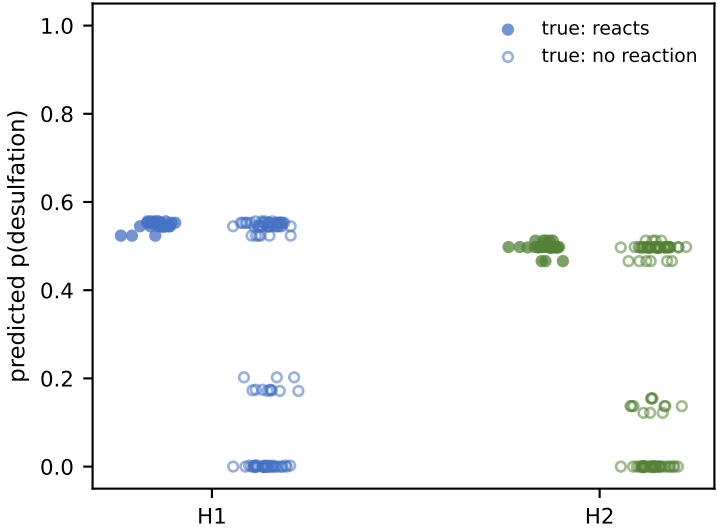
A Rules are recoverable from structure
(20-compound panel, held-out splits)



B Six sealed compounds, never seen in training
(H6 = DMSP negative control)

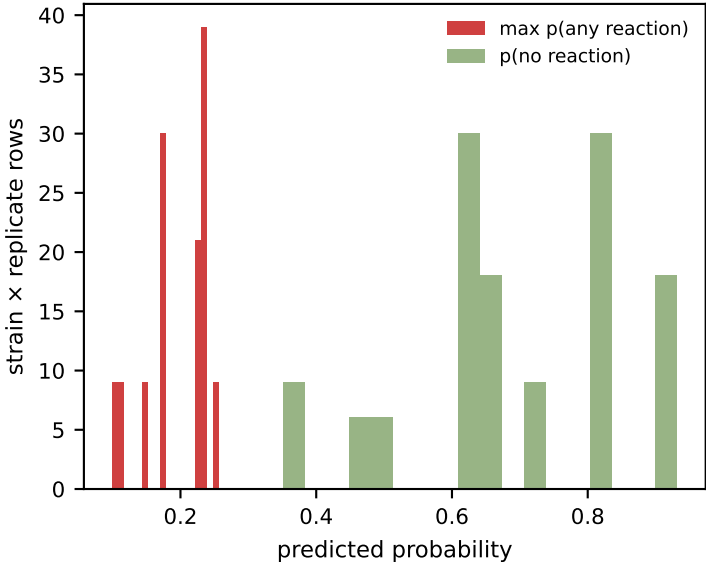


C Desulfation transfers to unseen sulfates
rule ROC-AUC 0.76 / 0.77 vs similarity 0.50

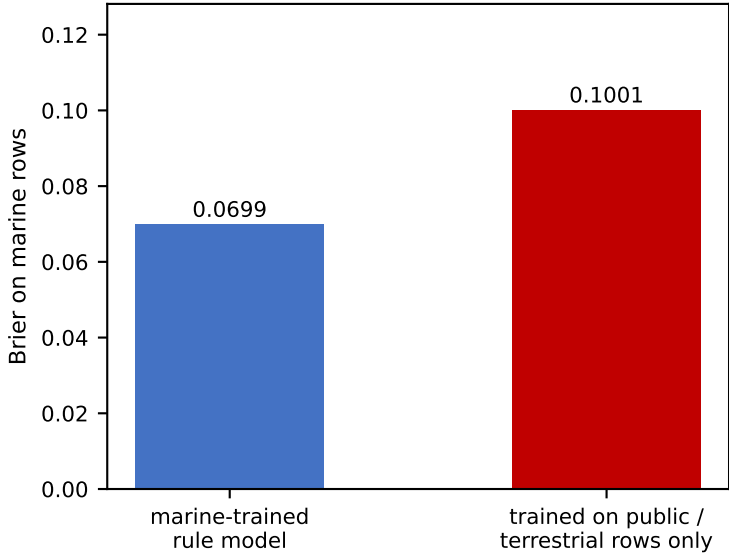


similarity baseline is constant per compound (ROC 0.50):
it cannot say WHICH strain reacts

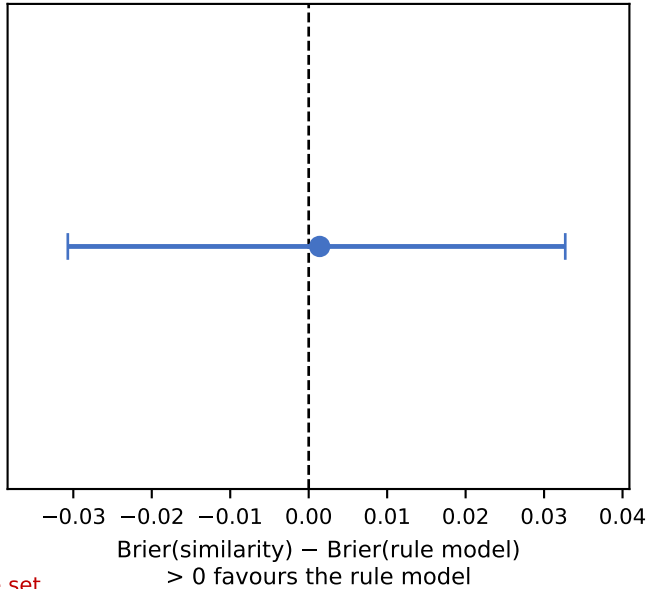
D Calibrated abstention on DMSP
(carries no trained substructure)



E Marine data advantage = 0.0302
(secondary success criterion)



F Underpowered at n = 6 sealed compounds
P(rule better) = 0.56, CI spans zero



SIMULATED DATA — labels are drawn by noisy-OR from the pre-registered rule priors, not measured. These panels demonstrate that the pipeline recovers a known rule set, calibrates its confidence and abstains correctly; they are an engineering positive control and carry NO biological claim. Panel F is the informative negative result.